LESSON PLAN

WEAPON GUIDANCE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	18
Total No. of Lesson Plans	22	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Definition of bonding, various types of bonding, bonding methods
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Definition of bonding, various types of bonding, bonding methods.
5. Specific Objectives	19.1.1.1Trainee will be able to explain aircraft wiring and Causes of wiring degradation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of bonding wire installation; Safety: Verify power off and lockout/tagout; Supervision: Instructor monitors tool usage and wire tension.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of bonding in aircraft structures?
(b) List two common causes of wiring degradation.
(c) Why is electrical continuity essential for aircraft safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of bonding • Purpose of bonding system • Static electricity hazards • Lightning strike protection • Corrosion prevention	L & Ds	PPT, IB	15 min
(b)	• Types of bonding: Structural vs Electrical • Types of bonding: Permanent vs Temporary • Types of bonding: Direct vs Indirect • Bonding wire specifications • Material selection criteria	L & Ds	PPT, PDF	20 min
(c)	• Bonding methods: Mechanical clamps • Bonding methods: Welding techniques • Bonding methods: Adhesive bonding • Inspection of bonding points • Maintenance procedures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between structural and electrical bonding.
(b) Describe the correct procedure for inspecting a bonding clamp.
(c) Identify three causes of wiring degradation related to poor bonding.

12. Home Assignment and Follow Up	Review LP 19 on connector identification and prepare a brief note on how proper connector marking supports effective bonding maintenance.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Various types of connectors their identification, marking of plugs and sockets.	Refer to the master syllabus reference list	LP 19

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min